

Figure 3. AGC remains diagnostic

Best-window AGC follows a partial bias-variance pattern, but boundary pressure prevents theorem-level closure.

Status: diagnostic; not a final quantitative theorem

A Original window-law fit

M4 AGC best-window law fits

basis	rho_exponent	sigma_a_exponent	r2	rmse_log10
hadamard_paired	-1.2265e-16	2.3494e-17	1	2.3738e-16
random_binary	-0.12418	-0.31575	0.55037	0.15503
random_uniform	-0.064788	-0.18087	0.43544	0.10676
srht_paired	-0.026909	-0.086348	0.29466	0.065111

B Targeted saturation

M4 targeted AGC best-window saturation

basis	lower_bound	interior	upper_bound	interior_frac	boundary_frac
hadamard_paired	25	20	0	0.44444	0.55556
random_binary	19	26	0	0.57778	0.42222
random_uniform	25	20	0	0.44444	0.55556
srht_paired	25	20	0	0.44444	0.55556

C Targeted fit

M4 targeted AGC window-law fits

basis	rho_exponent	sigma_a_exponent	r2	rmse_log10	n
hadamard_paired	-0.066542	-0.098988	0.70679	0.059692	45
random_binary	-0.29553	-0.64081	0.82086	0.20694	45
random_uniform	-0.23064	-0.48469	0.74867	0.19873	45
srht_paired	-0.17862	-0.33459	0.7866	0.13481	45

D Boundary-aware fit

M4 boundary-aware AGC window fits

basis	n_exact	n_upper_bounded	rho_exponent	sigma_a_exponent	hinge_rmse_log10	interval_satisfaction_frac
hadamard_paired	20	25	-0.076542	-0.11899	0.054044	0.4
random_binary	26	19	-0.43714	-1.0243	0.10055	0.8
random_uniform	20	25	-0.47896	-0.99675	0.028644	0.8
srht_paired	20	25	-0.3145	-0.59973	0.042365	0.64444

AGC window-law evidence across the original, targeted, and boundary-aware fits. The figure is intentionally labelled diagnostic because many best windows remain at the grid boundary.